

Lachlan Biersteker – Engineering Physics

[Portfolio](#) | 403-550-8730 | Lachlan.biersteker1@gmail.com | github.com/LachlanB56 |

SUMMARY

Deans list second-year Engineering Physics student at the University of British Columbia with hands-on experience in embedded systems, C/C++ programming, data analysis and machine learning in python, as well as PCB and CAD design. Seeking a 4 month internship to apply physics and programming fundamentals to impactful problems.

EDUCATION

University of British Columbia — Vancouver, BC

Sept 2025 – Present

B.A.Sc., Engineering Physics (Second Year) · Expected Graduation 2030

- Relevant coursework: Differential & Integral Calculus, Linear Algebra, Programming in C/C++, Quantum Mechanics, Thermodynamics & SHM, Electromagnetism, AC/DC Circuits.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, MATLAB/Simulink, SQL

Embedded & Hardware: STM32, Nordic, Arduino, I2C, SPI, UART

Design & CAD: Altium, SolidWorks, KiCad

Tools: VS Code, PlatformIO, Git/GitHub, STM32CubeIDE, IntelliJ IDEA

Certifications: Altium PCB design · IBM Basics of Quantum Information · MATLAB Onramp – MathWorks

EXPERIENCE & LEADERSHIP

UBC Aero Design Hardware Subteam

Sept 2026 - Current

- Hands on experience with embedded systems, circuit design, sensor integration, and hardware troubleshooting.
- Implementing embedded hardware around STM32 microcontrollers, integrating various avionic sensors.
- Designing and testing custom power distribution modules to safely supply the propulsion and avionics systems.
- Performing PCB bring-up, debugging, SMD soldering, and rework.

Coaching & Instructing in Athletics

Winter 2022 – Summer 2025

- Ski instructor - Training all ages & skill levels, enforcing safety protocols & adapting to dynamic conditions.
- Bike instructor - Instructed cycling and roadway safety, prioritizing safety getting 100+ kids biking on their own.
- Basketball Coach - Ran practices and coached games, developing players' skills and team strategy; leading the team to a first-place finish in the league.

TECHNICAL PROJECTS

Wearable Motion-Sensing System for Machine Learning Exercise Form Analysis

· [GitHub](#)

- Effectively built a wearable device that automatically detects reps in an exercise and tells user how they can correct their form for each rep using 2 machine learning models trained on hours of exercise and IMU data.
- Programmed microcontroller in C++, developed signal processing and machine learning in python achieving 85% segmentation accuracy for rep detection and 95% on form rating metrics.
- Fabricated and efficiently performed bring up of custom Altium designed compact PCB to stream data via Bluetooth.

Self Stabilizing Plane & STM32 Flight Motion Sensing System · [GitHub](#)

- Developed embedded firmware in C using STM32CubeIDE and HAL libraries to interface an IMU and barometric pressure sensor over I2C and UART as well as two servos as ailerons controlled using PID tuning logic.
- Acquired, processed, and debugged real-time sensor data over UART and implemented sensor fusion and complimentary filter to give clean values for roll pitch and yaw used in the PID controller.

SolidWorks Prototype Toothpaste Dispenser for Disabled Individuals · [GitHub](#)

- Collaborated with a team of classmates to design a toothpaste dispenser in SolidWorks for individuals with limited motor control for a school project.
- Conducted force analysis on the design to validate the design's robustness.
- Created a technical memo and delivered an oral presentation of the device's features to the class.